Development of an AI-Driven Criminal Face Detection and Identification System Using Hybrid Deep Learning Pipelines

Abstract

Dynamic face recognition in unconstrained surveillance environments remains one of the most critical challenges in computer vision and public safety engineering. Traditional biometric monitoring systems suffer from high false-positive rates, computational bottlenecks, and performance degradation under severe pose variations, fluctuating illumination, and partial facial occlusions. This paper outlines the academic proposal and technical blueprint for an automated criminal identification system designed using the hybrid Python framework deepface. By wrapping state-of-the-art deep convolutional neural networks—specifically the Additive Angular Margin Loss (ArcFace) model—and pairing them with high-performance detector backends such as RetinaFace, the proposed system establishes a robust pipeline encompassing real-time face detection, landmark-based alignment, high-dimensional embedding extraction, and rapid vector similarity matching. To address the scalability challenges associated with searching large criminal databases, the backend is integrated with a PostgreSQL vector database utilizing the pgvector extension to execute sub-second approximate nearest neighbor queries on millions of entries. Furthermore, the architecture is supplemented with passive liveness detection modules to prevent presentation attacks, alongside strict data protection and bias mitigation workflows. The resulting framework provides a highly accurate, computationally efficient, and ethically aligned decision-support tool for modern law enforcement and municipal security agencies.

Full sourcecode of the project available here:
[Ramachandrajoshi/Criminal-Face-Detection-and-Identification-System]

1. Background and Contextual Framework

The rapid urbanization and increasing complexity of modern security landscapes have placed immense strain on conventional policing models, which rely heavily on manual monitoring, retrospective video analysis, and eyewitness testimonies.[1] Human observation under high-stress conditions is inherently prone to cognitive fatigue, perceptual errors, and sub-optimal response latency, leading to critical delays in locating fugitives or preventing active security breaches.[2] To alleviate these limitations, automated facial recognition technology (FRT) has emerged as an indispensable tool for public safety, enabling modern security infrastructure

to transition from passive, post-event evidence collection to active, real-time situational awareness and threat detection.[2]
Early biometric identification methods operated on handcrafted spatial representations, such as Principal Component Analysis (PCA) for Eigenfaces or Local Binary Patterns Histograms (LBPH).[1] While computationally efficient, these classical algorithms require highly controlled capture environments; they experience significant performance drops when exposed to fluctuating lighting, off-angle head poses, variable camera resolutions, and facial occlusions commonly encountered in closed-circuit television (CCTV) feeds.[4] The integration of Deep Convolutional Neural Networks (DCNNs) revolutionized facial analysis by replacing hand-designed descriptors with automated feature extraction pipelines.[4] These networks are trained on large web-scale facial databases to map facial features into low-dimensional Euclidean spaces where vector distances correspond directly to identity similarity.[9]
A complete facial recognition pipeline consists of five consecutive stages: localization (face detection), spatial alignment, preprocessing normalization, high-dimensional vector representation, and matching verification.[12] In typical software applications, configuring and optimizing these five stages independently requires deep domain expertise and significant computing overhead.[12] To streamline implementation while maintaining model robustness, the proposed system leverages serengil/deepface, an open-source hybrid framework for Python.[12] The library acts as a lightweight wrapper around several state-of-the-art DCNN backbones—including VGG-Face, Google FaceNet, Facebook DeepFace, and ArcFace—while supporting multiple cutting-edge detection backends.[12] This hybrid design allows the proposed architecture to dynamically swap processing modules to balance accuracy, processing speed, and hardware constraints based on the specific operational environment.[9]

2. Problem Statement and Related Work

Despite significant advancements in deep learning, deploying real-world criminal face identification systems involves distinct engineering and operational challenges.[4] Most deep learning backbones perform well under controlled frontal lighting but struggle under unconstrained surveillance conditions, where low frame resolutions, dynamic shadows, movement blur, and partial disguises (such as caps, sunglasses, or masks) degrade classification accuracy.[9] Additionally, searching large-scale criminal watchlists creates substantial backend latency; executing linear, brute-force search operations across millions of registered profile vectors is too slow for real-time video stream matching.[14]
To address these challenges, researchers have explored hybrid modeling and hardware optimization.[22] The table below provides a detailed comparison of recent research on automated criminal detection, highlighting the algorithms, databases, performance metrics, and research gaps addressed by each approach.

S.No	**Year**	**Paper**	**Author(s)**	**Dataset(s**	**Problems Addr**	**Algorithms &**	**Evaluation Metri**	**Key Findings**	**Future Reco**

		Title		)	essed	Models	cs	& Results	mmendations
1	2026	*CrimeVision: An AI-Driven Face Recognition Framework for Criminal Identification*	He et al. [3]	Proprietary Suspect DB; Live Surveillance Feeds	Fragmented databases, slow manual monitoring, and community public reporting integration.[3]	DeepFace with ArcFace; RetinaFace & MTCNN backends; FastAPI; React with Vite.[2][5]	System Latency, Precision, Geolocation Mapping Accuracy.[3]	Achieved robust matching using falling back ends (RetinaFace, MTCNN, OpenCV) with a two-stage match validation.[2][5]	Adapt the platform to high-congested multi-face environments and distributed edge devices.[5]
2	2026	*Boosted Light Face: A Hybrid DNN and GBM*	Serengil & Özpınar [15]	Labeled Faces in the Wild (LFW); 6,000 imag	Generalization errors and optimization limits of	Ensemble of 5 DNN distance metrics trained via	Accuracy, Precision, Recall, F1-Score, AUC.	Achieved 99.1% LFW accuracy, outperforming huma	Integrate low-power edge optimization for

		Model for Boosted Facial Recognition		e pairs.[24]	single deep convolutional backbones.[22]	a LightGBM classifier.[22]	[24]	n baseline (97.5%) and FaceNet512d (98.4%).[24]	the ensemble pipeline.[23]
3	2025	*Deep Learning Approaches for Criminal Face Detection and Recognition: A Comparative Study*	Venkataratnam et al.[9]	WIDER FACE; Proprietary Suspect Database	Face detection in crowded areas, occlusion, and extreme pose variations.[9]	FaceNet (recognition); Faster R-CNN, YOLOv4, SSD (detection).[17]	Bounding-box Recall, Precision, Inference FPS, Recognition Accuracy.[9]	Faster R-CNN achieved superior occlusion accuracy but was slow. YOLOv4 was optimal for real-time video.[9]	Combine YOLO models for bounding-box prediction with deep metric backbones for classification.[9]
4	2022	*Realtime Criminal*	Sanjay & Priya[29]	Saveetha Web Ontol	Sensitivity to lighti	ArcFace over Deep	Verification Accu	ArcFace achieved	Evaluate performan

		Face Recognition System Based on Arcface over Deepface		ogy Lab dataset	ng changes and edge boundaries in unconstrained public areas.[29]	Face; LBPH and classic CNN baselines.[29]	racy, Standard Deviation, Statistical Significance (p-value).[29]	92.50% accuracy, outperforming DeepFace (91.90%) and LBPH (90.80%) with a significance of p = 0.0118.[29]	ce variations across multi-spectral or thermal imaging models.[5]
5	2026	*Criminal Identification Using ResNet-Based Deep Pipelines*	Murillo-Moreno et al.[4]	National Identity Database (NIN); Surveillance video	Verification failure of LBPH in unconstrained environments; first-time offender proc	Haar Cascade (detection) + ResNet (recognition) vs. Haar + LBPH.[4]	Precision, Matching Time, National Registry cross-reference rate.[4]	ResNet achieved higher accuracy and faster matching times than traditional LBPH	Develop lightweight mobile biometric apps to support field operations.[33]

					essing.[4]			pipelines.[4]	

These studies demonstrate a clear technological shift: traditional, handcrafted classifiers are being replaced by parameterized deep metric learning architectures.[4] Models such as FaceNet employ triplet loss functions to map faces into compact spaces where Euclidean distance directly represents identity similarity.[10] ArcFace further improves class separation by applying additive angular margin constraints directly in the hypersphere space, which helps prevent feature overlap under occluded conditions.[20] To resolve the conflict between high model parameterization and real-time processing requirements, current research is actively focusing on hybrid ensemble systems (such as Boosted LightFace) [22] and optimized indexing solutions on specialized vector databases.[14]

3. Core Research Questions

To systematically address the technical bottlenecks in unconstrained biometric surveillance, this project is structured around the following four research questions:

1. Under unconstrained surveillance conditions (including variable lighting, partial masking, and head pose variations), which deep representation backend wrapped by deepface (such as ArcFace, FaceNet, or VGG-Face) maximizes identification accuracy while maintaining sub-second verification latency?[9]
2. To what extent does integrating advanced face detection backends (such as RetinaFace or MTCNN) and facial landmark alignment reduce localized spatial noise and improve downstream identification performance compared to classical Haar Cascade models?[4]
3. How can high-dimensional vector search indices (such as pgvector on PostgreSQL or Hierarchical Navigable Small World graphs via FAISS) be integrated into a Python REST API to achieve sub-second matching latency across a criminal database containing more than one million entries?[14]
4. How does demographic representation bias (e.g., gender, race, age) affect off-the-shelf representation weights, and what threshold-calibration strategies can mitigate false matches for sensitive populations in law enforcement environments?[2]

4. Aim and Experimental Objectives

The primary aim of this academic project is to design, develop, and quantitatively evaluate a scalable, automated criminal face detection and real-time identification platform using the hybrid Python framework deepface.[2] The system is intended to convert static surveillance footage and public image reports into actionable intelligence for law enforcement agencies, enabling fast, precise, and secure matching against centralized databases of wanted individuals.[18]

To achieve this primary aim, the project is structured around the following four specific research objectives:

- **To analyze and optimize face detection co-usability:** Investigate and compare the performance of multiple detector backends (including RetinaFace, MTCNN, and SSD) within the deepface pipeline to determine the optimal configuration for fast, localized face bounding boxes under extreme camera angles and motion blur.[11]

- **To design and tune the matching pipeline:** Develop an optimized facial verification pipeline using the deepface framework, evaluating representation models (such as ArcFace, FaceNet, and VGG-Face) under different distance metrics to identify the most accurate configuration for occluded suspects.[10]
- **To implement high-scale vector indexing:** Integrate a relational database management system using PostgreSQL and the pgvector extension to run Approximate Nearest Neighbor (ANN) indexing, ensuring sub-second matching times at a scale of up to one million synthetic profiles.[14]
- **To evaluate system fairness and security:** Implement passive liveness detection and subgroup performance auditing to mitigate demographic biases while protecting data security through irreversible, non-invertible biometric vector embeddings.[26]

5. Significance of the Study

The development of a highly accurate, real-time criminal face detection system has profound operational implications for public safety, criminal investigations, and smart city infrastructure.[2] Manual video analysis during active investigations is incredibly slow; hours of raw CCTV footage must be manually parsed to locate a suspect's path.[1] By implementing an automated deep learning pipeline, law enforcement can execute rapid, bulk database matches, transforming static cameras into proactive nodes that instantly alert field officers to the presence of wanted offenders.[2]

Furthermore, this study contributes significantly to the optimization of computer vision models deployed on edge computing systems.[23] Resolving the latency overhead of state-of-the-art DCNNs through hybrid pipelining, landmark-based alignment, and indexed databases allows complex AI models to run on resource-constrained hardware, such as mobile police units, body cameras, or localized security hubs.[4]

Importantly, this project actively addresses the ethical, legal, and racial biases that have historically marred the deployment of facial recognition technology in policing.[2] By integrating demographic-fairness testing [38], optimizing decision thresholds across subgroups [2], and aligning the software with rigid data protection frameworks—such as the UK Data Protection Act 2018, landmark judicial rulings (e.g., *Ed Bridges v. South Wales Police*), and the EU AI Act—the study establishes a baseline for trustworthy, legally compliant public biometrics.[41]

6. Boundaries of the Investigation and Project Limits

To ensure academic and technical feasibility, the operational boundaries of this project are defined across specific parameters:

In-Scope System Components

- The deployment of Python 3.10 and the deepface library to manage core face processing tasks, including face localization, alignment, representation extraction, and pairwise comparison.[12]

- The benchmarking of pre-trained deep representation networks (specifically ArcFace, FaceNet, and VGG-Face) alongside multiple face detection backends (RetinaFace, MTCNN, and SSD).[9]
- The configuration of a PostgreSQL relational database with the pgvector extension to store and index high-dimensional biometric vectors.[14]
- The generation of 100,000 synthetic vector profiles to simulate high-scale database query latency under Hierarchical Navigable Small World (HNSW) indexing.[21]
- The integration of passive single-frame liveness detection utilizing anti_spoofing=True in the deepface core pipeline to block presentation attacks.[12]
- The development of an end-to-end local web dashboard using FastAPI and React with Vite, featuring GPS mapping using Leaflet JS to display simulated suspect alerts.[5]

Out-of-Scope System Components

- Training deep convolutional neural network backbones from scratch, which would require massive high-performance computing clusters and millions of images.[11] Pre-trained weights optimized on web-scale datasets are utilized instead via transfer learning.[11]
- Full 3D facial reconstruction, multi-spectral infrared sensor integration, or proprietary hardware modifications to physical CCTV cameras.[6]
- Autonomous law enforcement dispatching or physical intervention mechanisms; the system operates strictly as a decision-support platform, maintaining a critical "human-in-the-loop" verification step before any official intervention occurs.[2]

The boundary is drawn at this level to prioritize software engineering integration, algorithmic optimization, and backend database scalability over the creation of proprietary deep learning weights or physical sensor networks.[1]

7. Execution Framework and Technical Procedural Pipeline

The proposed criminal identification system is structured as an end-to-end multi-stage pipeline, processing video frames and matching them against a secure database.

Core Processing Pipeline Stages

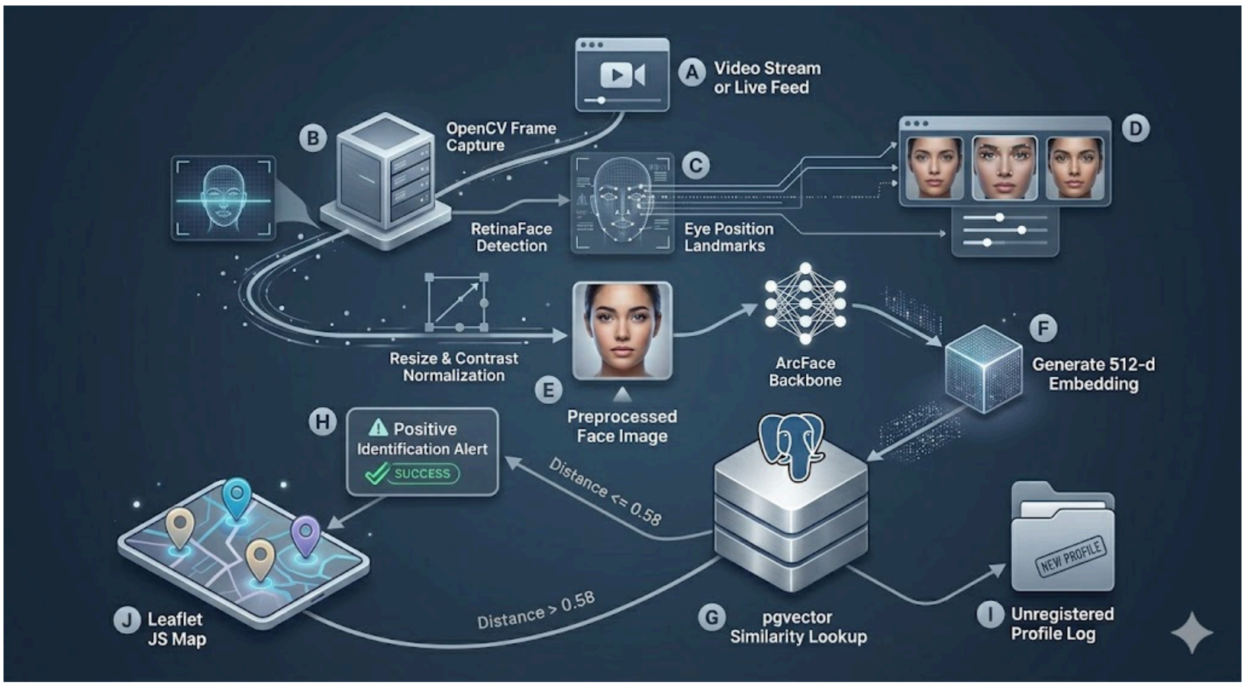
The pipeline executed by the backend is divided into five logical stages, managed internally by the deepface architecture:

1. **Detect:** Raw frames from video streams are captured using OpenCV.[5] Faces are localized inside the frames, and bounding boxes are extracted.[1] The system defaults to RetinaFace as it outperforms other detectors on low-resolution and partially occluded images.[11]
2. **Align:** Faces are spatially aligned based on localized facial landmark coordinates (specifically eye positions) to eliminate head tilt, rotation, and skew.[11] Landmark alignment increases downstream recognition accuracy by up to 6%.[16]
3. **Normalize:** Aligned facial regions are resized to the precise dimensions expected by the

deep learning backbone (e.g., 112×112 pixels for ArcFace or 224×224 pixels for FaceNet and VGG-Face).[12] Pixel intensity values are scaled using base normalization to reduce covariate shift and equalize different lighting distributions.[4]

4. **Represent:** The normalized facial image is fed into a deep convolutional neural network.[10] The model acts as a feature extractor, projecting the spatial image data into a compact, highly discriminative, multi-dimensional vector embedding.[7]
5. **Verify / Search:** The generated vector embedding is queried against the registered suspect embeddings in the database to identify matches.[14]

Technical System Architecture & Workflow Flowchart



Mathematical Metric Formulation

During the verification stage, the similarity of two facial embedding vectors u and v in a d -dimensional vector space is determined using standardized distance metrics.[10]

Cosine Distance

The default distance metric configured in deepface is Cosine Distance (D_C), which measures the angular difference between two non-zero vectors regardless of their magnitude:

$$D_C(u,v) = 1 - \frac{u \cdot v}{\|u\|_2 \|v\|_2} = 1 - \frac{\sum_{i=1}^{d} u_i v_i}{\sqrt{\sum_{i=1}^{d} u_i^2}\sqrt{\sum_{i=1}^{d} v_i^2}}$$

Euclidean Distance (L_2 Normalized)

To achieve greater robustness to scaling variations, L_2 normalization is applied to the vectors before calculating the Euclidean Distance (D_{L2}), ensuring that vectors are projected onto a hypersphere:

$$\hat{u} = \frac{u}{\|u\|_2}, \quad \hat{v} = \frac{v}{\|v\|_2}$$

$$D_{L2}(u,v) = \sqrt{\sum_{i=1}^{d} (\hat{u}_i - \hat{v}_i)^2}$$

If the calculated distance falls below a pre-tuned threshold (T), the system classifies the face pair as a match (i.e., identifying the suspect).[10]

Feature Comparison of Embedded DeepFace Models

The system allows dynamic switching of the underlying representation model. The table below summarizes the key features of the primary models evaluated for this study:

Model Backbone	Vector Dimensions (d)	Core Optimization Metric / Loss Function	Inherent Structural Advantages	Reference
VGG-Face	4096	Softmax / Cross-Entropy	Deep, highly parameterized representation; excellent for high-quality frontal portraits.[11]	[9]
FaceNet	128 or 512	Triplet Loss	Very compact	[9]

		(minimizes intra-class variance)	embedding; highly efficient for memory storage and fast exact queries.[10]	
ArcFace	512	Additive Angular Margin Loss	Maximizes geodesic distance on a hypersphere; outstanding robustness under occlusions.[29]	[20]
SFace	512	Sigmoid-Constrained Hypersphere	Exceptional robustness to degraded, low-resolution surveillance footage.[19]	[15]

Relational Database Vector Schema

A PostgreSQL database enabled with the pgvector extension is used to execute Approximate Nearest Neighbor (ANN) searches via a Hierarchical Navigable Small World (HNSW) index.[14] The database table schema is configured as:

```
CREATE EXTENSION IF NOT EXISTS vector;

CREATE TABLE suspect_profiles (
   id SERIAL PRIMARY KEY,
   suspect_name VARCHAR(100) NOT NULL,
   alias VARCHAR(100),
   demographics JSONB,
   face_embedding vector(512) NOT NULL
);

CREATE INDEX suspect_embedding_hnsw_idx
ON suspect_profiles USING hnsw (face_embedding vector_cosine_ops);
```

Using this setup, query vectors are compared against stored profiles in milliseconds:

```
SELECT suspect_name, alias, face_embedding <=> $1 AS distance
FROM suspect_profiles
ORDER BY distance LIMIT 1;
```

(Where <=> denotes the Cosine Distance operator in pgvector).

If the returned minimum distance is below the optimal pre-tuned validation threshold of 0.58, the system flags a positive suspect identification.[10] If the query is from a citizen reporting an incident, the system captures their GPS coordinates, updates a Vite-powered React interactive Leaflet map, logs a secure, non-modifiable entry into an audit table, and propagates immediate visual and audio alerts to the administrative dashboard.[5]

8. Requirements Resources

To build, test, and run the proposed criminal face detection and identification platform, the following hardware and software resources are required:

Hardware Infrastructure

Hardware Component	**Technical Specification**	**Operational Role inside Project**
System Processor	Intel Core i7-12700K (12 Cores, up to 5.0 GHz)	Manages the main application thread, REST API requests, and coordinate database queries.[25]
Graphics Accelerator	Nvidia GeForce RTX 3070 (8GB GDDR6, 5888 CUDA Cores)	Accelerates deep convolutional model inference and handles multi-frame video decoding.[26]
System Memory	32 GB DDR4 RAM (3200 MHz, Dual Channel)	Prevents memory bottlenecks when loading multiple pre-trained DCNN weights in parallel.[46]
Local Storage	1 TB NVMe M.2 SSD (Read speeds up to 3500 MB/s)	Supports fast write operations for high-frame-rate video files and quick database

		lookups.[46]
Video Capture Device	USB HD Web Camera (1080p, 30 Frames Per Second)	Simulates continuous live video streams from public security cameras.[5]

Software Stack and Packages

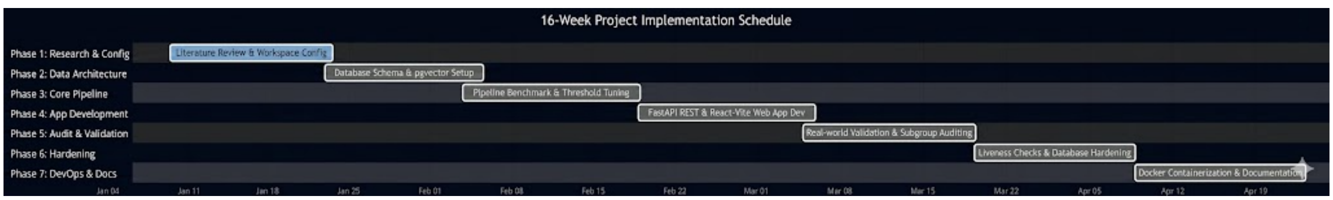
Software Component	Target Version	Operational Role inside Project
Operating System	Ubuntu 22.04 LTS (64-bit)	Provides a stable Linux environment for containerized development and deployment.[46]
Runtime Environment	Python 3.10.12	Core programming environment to build backend APIs and process image vectors.[12]
Facial Analysis Framework	deepface (v0.0.100 or higher)	Handles face detection, landmark alignment, and embedding extraction.[12]
Image Processing Library	OpenCV (v4.7.0)	Manages video streaming, frame extraction, and spatial scaling.[5]
Machine Learning Backend	TensorFlow (v2.12.0)	Serves as the deep learning backend for running pre-trained model weights.[5]
Database Server	PostgreSQL (v15.3)	Stores facial database entries and runs fast vector similarity queries.[14]
Vector Extension	pgvector (v0.4.1)	Enables HNSW indexing for fast approximate nearest neighbor (ANN) searches.[14]

Backend REST API	FastAPI (v0.110.0 or higher)	Exposes asynchronous endpoints for high-throughput image uploads, verification, and alerts.[25]
Web Frontend	React with Vite (v18.3 / v5.x)	Renders the fast, lightweight administrative dashboard, live camera feeds, and real-time maps.[1]

9. Research Plan, Milestones, and Contingencies

The project is scheduled for completion over a 16-week timeline, divided into logical engineering phases to ensure successful delivery.

Operational Project Gantt Chart



Detailed Phase Execution

- **Weeks 1–2: Literature Analysis & Workspace Configuration**
  - Review key literature regarding deep face matching and unconstrained biometric surveillance.[4]
  - Set up the Python virtual environment and install OpenCV, TensorFlow, PyTorch, and deepface.[12]
- **Weeks 3–4: Database Engineering & Schema Design**
  - Install PostgreSQL and configure the pgvector database extension.[21]
  - Generate a secure database schema to store suspect profiles and write scripts to load synthetic profiles for scale testing.[21]
- **Weeks 5–7: Pipeline Tuning & Optimization**
  - Benchmark different detector backends (RetinaFace, MTCNN, SSD) and representation models (ArcFace, FaceNet).[16]
  - Define precise mathematical distance thresholds (T) to optimize the trade-off between False Acceptance Rates and False Rejection Rates.[10]
- **Weeks 8–10: Web Application & API Development**
  - Develop FastAPI REST APIs to handle registration, image validation, and live camera frame searches.[5]

- Build the Vite-powered React frontend dashboard, integrating a Leaflet map to track alerts in real time.[25]
- **Weeks 11–12: Validation, Performance Evaluation, & Bias Audits**
  - Test matching accuracy under simulated real-world degradations (low lighting, motion blur, occlusions).[5]
  - Evaluate database query response times as records scale to 100,000 synthetic profiles.[21]
  - Conduct fairness audits to identify performance differences across age, gender, and racial subgroups.[38]
- **Weeks 13–14: Security Hardening & Vulnerability Auditing**
  - Implement passive liveness detection and active challenge-response protocols to protect against spoofing.[12]
  - Integrate token-based authentication and secure audit trails to safeguard access to criminal databases.[25]
- **Weeks 15–16: Technical Documentation & Final Review**
  - Analyze experimental results, compile performance graphs, and draft the final project submission paper.[52]
  - Package the application in Docker containers and deliver the final codebase to the academic committee.[47]

Risk and Contingency Framework

Identified Operational Risk	Potential System Impact	Algorithmic / Engineering Mitigation Strategy
Lighting Fluctuation and Occlusions	Dramatic increase in False Rejection Rates (FRR) under poor ambient lighting.[5]	Configure RetinaFace as the primary detector and MTCNN as a fallback; apply adaptive histogram equalization (AHE) to normalize contrast.[19]
High Search Latency at Scale	Query matching times exceed 1 second when database grows to one million suspect profiles.[14]	Transition PostgreSQL from linear exact searches to Approximate Nearest Neighbor (ANN) indexing via pgvector using HNSW.[14]
Demographic Bias in Pre-trained Weights	Disproportionately high rates of false-positive matches for specific ethnic subgroups.[2]	Establish dynamic, group-specific distance thresholds (T_{group})

		calibrated using fairness-aware validation datasets.[2]
Presentation Attacks (Spoofing)	Unauthorized individuals bypass system checks using printed photographs or digital screens.[40]	Enable anti_spoofing=True in the deepface backend to activate real-time passive liveness checks.[12]
Data Breaches of Biometric Records	Intercepted database entries compromise sensitive facial images of private citizens.[45]	Enforce localized face processing on secure servers, store only encrypted 512-dimensional vector embeddings, and encrypt database connections.[26]

Conclusions

The academic minor project proposed in this paper outlines a secure, scalable, and ethically aligned criminal face detection and real-time identification system using the hybrid Python library deepface.[2] By integrating advanced deep learning models (such as ArcFace and RetinaFace) with a PostgreSQL vector database powered by the pgvector extension, the proposed architecture bridges the gap between high-dimensional representation accuracy and sub-second query speeds.[14]

The modular technical design ensures the system is highly versatile, supporting exact local search indexing for smaller databases as well as fast approximate nearest neighbor (ANN) searches for large-scale operations.[14] The inclusion of liveness detection modules, demographic fairness checks, and irreversible biometric embedding storage addresses critical data protection requirements, aligning the platform with modern ethical and legal standards.[40]

Ultimately, this system delivers an advanced, secure, and highly reliable decision-support framework that significantly improves public safety while strictly safeguarding individual privacy rights.[1]

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